Customer wants an immediate spectroscopy method and sensor configuration for occasional off-spec events in a liquid process with a suspected, unnamed contaminant, and asks what information and first experiment must come first.

DECISION STATUS **Target missing**

No method yet: the target must be defined first; methods are screening directions only.

Fluorescence, fluorimetric probe · 20/36

- Preference is soft; a lab interface stays valid for the feasibility experiment. REF §3 R_Q3C_05
- Documented Method-interface edge (S01 Table 1). XLSX S01 p4-6 R_Q2_01

Not possible yet: No exact sensor can be named before the target is defined. R_Q1_01 REF §3 H05

Recommended methods

#1
20 /36
range [4–31]

Fluorescence · fluorimetric probe

Confidence: low | ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	5 /9	Q3	7 /9
Q4	0 /3		

#2
20 /36
range [5–30]

MIR · ATR probe

Confidence: low | ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	5 /9	Q3	7 /9
Q4	0 /3		

#3
20 /36
range [4–30]

NIR · diffuse reflection probe

Confidence: low | ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	5 /9	Q3	7 /9
Q4	0 /3		

▼ More methods (2)

#4
20 /36
range [4–30]

Raman · Raman probe

Confidence: low | ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	5 /9	Q3	7 /9
Q4	0 /3		

#5

20 /36

range [5–31]

UV-Vis · ATR probe

Confidence: low | ⚠ critical question open

Q1	4 /9	Q1b	4 /9
Q2	5 /9	Q3	7 /9
Q4	0 /3		

Questions for the customer

1. high `MeasurementTask · target definition`

   What exactly is off-spec / what should the measurement decide? Which substance or property defines it?

   ▼ Why

   Without a defined target no comparable claims can be formed (spec 06 §6, H05).

   `R_Q1_01`

2. high `Sample · optical properties · fluorescence`

   Does the matrix fluoresce (known fluorescent components, colour)?

   ▼ Why

   if fluorescence=absent: leader Raman

   `R_Q1_R02`

3. high `Sample · optical properties · opacity`

   Is the sample clear, turbid or opaque at the measuring point, and does it change?

   ▼ Why

   if opacity=clear: leader MIR

   `R_Q2_04`

To improve accuracy, add the customer's answers to the text above and click Analyze again.

Ruled out (1)

UV_VIS · UV_LENS
This interface collects light emitted by the sample itself; the sample is not a light source. `R_Q2_02`

Decision path · factors of the leading candidate per stage, with source

Q1

UNKNOWN **0** Invalid old calibration does not exclude the physical method; a new calibration plan is mandatory. `REF D01 risks, §5` `S01 p10` R_Q1_06

UNKNOWN **0** Independent validation material must be established, not assumed.
`REF D01/D02 experiment` `S01 p5,10` R_Q1_07

UNKNOWN **0** Without a fluorophore the direct concentration is a new hypothesis, not a fact.
`S01 p5,16` `REF §5` `H03` R_Q1_F01

Q1b

UNKNOWN **0** LOD/LOQ only from a scoped CapabilityProfile; the materials give none.
`spec 04 §1` `REF §3,5` R_Q1B_02

UNKNOWN **0** Achieved error vs allowed error needs a comparable metric; none in the materials.
`REF D01/D02 experiment` `spec 04` R_Q1B_04

UNKNOWN **0** A laboratory is not a known reference uncertainty. `REF D01,D02` `H01` R_Q1B_05

UNKNOWN **0** End-to-end latency needs a profile; spectral exposure alone is not the latency.
`spec 04 §1` R_Q1B_06

UNKNOWN **0** Both range ends must be inside the calibration/detector scope.
`REF D01,D02` `S01 p16,18` R_Q1B_08

UNKNOWN **0** Qualitative sensitivity rating: context only, never a score. `XLSX sensitivity + notes`
R_Q1B_10

Q2

MATCH **+1** Documented Method-interface edge (S01 Table 1). `XLSX` `S01 p4-6` R_Q2_01

UNKNOWN **0** A form missing from the list is UNKNOWN; MISMATCH only for a constructive conflict (solid vs liquid-only cell). `XLSX` `CL10` R_Q2_02

UNKNOWN **0** Excludes only this pair; other interfaces stay. Taking a sample out is R_Q2_10 (one cause, one rule). `S01 p5,10` R_Q2_03

UNKNOWN **0** A surface or local signal is not the bulk composition of a heterogeneous stream without a sampling plan. `S01 p8,13-14,17-18` `REF D08` R_Q2_07

UNKNOWN **0** Particle/bubble/surface variation must be covered by a sampling or averaging plan. Our recommendation to average is not a plan (no MATCH from it). `REF D01,D02` `S01 p15,17` R_Q2_08

Q3

UNKNOWN **0** No numeric limit in the materials. Hot extrusion does not exclude MIR.
`S02 p14` `REF D07` R_Q3C_01

UNKNOWN **0** Fouling + contact interface needs a feasible cleaning/maintenance plan; a promise is not a match. `REF D02,D07` R_Q3C_04

MATCH **+2** Preference is soft; a lab interface stays valid for the feasibility experiment. `REF §3` R_Q3C_05

UNKNOWN **0** Do not invent certification. `design comparator` R_Q3C_06

UNKNOWN **0** Long silica cables are documented; no universal allowed length. `S02 p6-10,18` `S01 p15` R_Q3C_07

MATCH **+1** Generic family is a soft concept check; a real cable with proven mismatch is R_Q3P_03. `S02 p6` `REF §5` R_Q3C_08

Risks and assumptions

high The customer's request for an exact sensor configuration now cannot be met: with an unnamed contaminant no sensor can be specified yet. What still meets the goal: a short lab screening of normal against off-spec samples with fibre probes, then a configuration chosen from the result.

high Ranking vs assessment: the ranking is empty, while the Method B assessment lists screening directions (fluorescence, MIR, NIR, Raman, UV-Vis, all about 20/36) with wide intervals, low confidence and no supporting evidence (Q4 = 0). The empty ranking is the better-supported side; the listed methods are screening directions only, and a spectral difference alone does not validate a calibration.

high Fluorescence and MIR score the same as the others and each has a wide score interval; the decisive unknowns (fluorescence of the matrix, opacity, contaminant identity) could change the leader. Keep any recommendation conditional.

medium Unknown contaminant: a method can show that a sample differs without showing what the contaminant is. A first model may only classify 'different from normal' and may react to harmless variation (batch, temperature, turbidity).

medium Occasional events mean few off-spec samples. Calibration and independent validation may be statistically weak, and reference analyses of past events may be missing or not representative.

medium Process conditions are unknown (temperature, pressure, fouling, water content, hazardous area, fibre distance). ATR crystals and probe materials need checking against temperature and chemistry, MIR fibre links are shorter and costlier than silica links, and fouling or turbidity can distort fluorescence and MIR signals.

medium Water content is unknown: strong water absorption could limit MIR sensitivity in aqueous liquids, and a matrix fluorescence background could mask a fluorescence signal.

First feasibility experiment

Define the target and show whether normal and off-spec liquid samples differ spectrally in a way that can be tied to an independently identified cause, before any in-line configuration is fixed.

Samples

To be agreed with the customer. Proposal: as many characterised off-spec samples as exist (ideally from several separate events) plus normal samples from at least 3 independent production batches, split into calibration and independent validation sets. — Collect normal samples across the normal range (batches, recipe or raw material lots, temperature, turbidity). Collect off-spec samples with the date, batch and process state of each event. Once the contaminant is identified, add spiked normal samples at several levels, including at least one spike series that breaks correlation with other process changes (e.g. temperature or batch). Keep a strict calibration/validation split by batch.

Reference method

Independent lab identification of the contaminant or quality problem (the customer's lab method). If none exists, options are GC/LC with MS identification, elemental analysis, or an established customer quality assay; the customer chooses.

Methods compared

Fluorescence, MIR

Interferents

normal batch-to-batch variation of the matrix, temperature changes, turbidity, particles or bubbles, possible matrix fluorescence or colour, water content (unknown)

Duration

About 4–8 weeks for the lab phase; longer if off-spec events are rare and fresh samples must be waited for. An in-line pilot would follow afterwards and take longer, especially under GMP change control.

1. Clarify with the customer what defines off-spec, what decision the measurement supports and the required response time.
2. Collect and document representative normal and off-spec samples (retained samples and fresh ones at the next event) with batch and process data.
3. Have the off-spec samples analysed by the independent reference lab to identify the contaminant or a measurable proxy attribute.
4. Measure all samples in the lab with fibre probes for fluorescence (fluorimetric probe) and MIR (ATR probe and, if the liquid allows, transflection probe) at controlled temperature, and compare each with the reference result.
5. Add spiked and designed samples once the target is known, including series that break correlations between contaminant and normal process variation.
6. Build calibration/classification models on at least two calibration batches and test them on independent validation batches; compare results of both methods against the reference.
7. Review results with the customer and agree on the next stage: a pilot in a bypass or sampling loop, or a revised target.

Success criterion: Agreed with the customer after the target is defined. Proposed form: off-spec and normal validation samples are separated by at least one method at an agreed miss and false-alarm rate. If a named contaminant is quantified, the validation error (RMSEP) is within the agreed limit and the bias against the reference is not significantly different from zero and within the agreed limit.

Parameters and attributes · 38 Facts

Facts

Value · unit · status. Hover a row for the customer's words. Edit → Recalculate.

Fact	Value	Unit	Status
MeasurementTask · measurement goal	—		unknown
MeasurementTask · target	contaminant		reported „…“
MeasurementTask · target definition	undefined		reported „…“
MeasurementTask · spatial scope	—		unknown
MeasurementTask · sampling plan	—		unknown
MeasurementTask · reference method	—		unknown
MeasurementTask · performance requirements	—		unknown
MeasurementTask · time requirements	—		unknown
MeasurementTask · target range	—		unknown
MeasurementTask · business goal	—		unknown
Sample · matrix	—		unknown
Sample · phase	liquid		reported „…“
Sample · physical form	—		unknown
Sample · optical properties · opacity	—		unknown
Sample · optical properties · fluorescence	—		unknown
Sample · heterogeneity	—		unknown

Fact	Value	Unit	Status
Sample · variability · formulation	—		unknown
Sample · variability · physical	—		unknown
Sample · sample state constraints · contact	—		unknown
Sample · sample state constraints · sampling	—		unknown
Sample · relevant components · fluorophore	—		unknown
Sample · relevant components · absorber	—		unknown
ProcessContext · deployment preference	—		unknown
ProcessContext · temperature range	—		unknown
ProcessContext · temperature c	—	degC	unknown
ProcessContext · operation constraints · fouling	—		unknown
ProcessContext · operation constraints · hazardous	—		unknown
ProcessContext · instrument distance	—		unknown
ProcessContext · assets · fiber	—		unknown
ProcessContext · assets · fiber profile	—		unknown
ProcessContext · assets · must reuse	—		unknown
ProcessContext · assets · calibration scope	—		unknown
MeasurementTask · reference method · independent	—		unknown
ProcessContext · assets · method	—		unknown

Fact	Value	Unit	Status
CustomerRequest · justification	—		unknown
CustomerRequest · proxy allowed	—		unknown
CustomerRequest · asks	exact_config_now		reported „…“
CustomerRequest · combination to keep calibration	—		unknown